A Scoped Empirical Claim on Magic–Learnability Coupling in θ-Quench Schwinger Dynamics

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Abstract

This note reports evidence for a conditional claim: in a clearly specified post-quench corridor, higher stabilizer-Renyi magic is associated with worse NNQS learnability (higher validation NLL), including after entropy control. The testbed is the $(1+1)$D open-boundary lattice Schwinger model with exact small-N dynamics ($N \lesssim 12$), supplemented by cross-family stress tests (TFIM, XXZ, ANNNI). We compute Loschmidt diagnostics, correlators, entanglement, and magic M_α, then evaluate learnability across GRU, MADE, RBM, and independent baselines with bootstrap/permutation/FDR gates. Additional checkpoint-GPU replications on Hyak strengthen the baseline cross-architecture evidence (GRU, MADE, RBM), support only the entropy-controlled endpoint in the current exact $N = 10$ rerun, and sharpen the regime boundary by marking heavy-mass and strong-coupling alternates as mixed/unsupported. The data support a robust entropy-controlled slope law in the tested corridor and cross-family cells, while strict all-regime and theorem-level universal statements are not supported.

1 Introduction

The central question is whether non-stabilizerness (measured by stabilizer-Renyi magic) predicts neural learnability degradation in real-time many-body dynamics. The goal is not a universal theorem claim; instead, the goal is a reviewer-safe empirical statement with explicit support boundaries and explicit failure boundaries.

Motivation and attribution. The motivating project direction for this repository came directly from the Institute for Nuclear Theory (INT) INTURN 26-3 project theme on quantum magic in topological vacuum dynamics. This note should therefore be read as an independent undergraduate implementation and exploratory extension of that project prompt and the surrounding literature, not as the origin of the underlying research question itself.

2 Primary claim and boundaries

Supported claim. In the declared dynamical corridor of this study, higher magic predicts higher validation NLL under entropy control, with positive lower confidence bounds for the magic slope in cross-family cell-level tests.

Not claimed. A model-independent globally valid law, strict all-regime positivity, and theorem-level guarantees are not established by this dataset.

Publication position. The current evidence supports a scoped mechanism claim suitable for a focused empirical paper (conditional law with boundary map), not a broad-law paper.

3 Related work

Our setup combines three strands: (i) Schwinger-model quench diagnostics (Loschmidt and correlator structure), (ii) stabilizer-magic complexity measures in many-body states, and (iii) neural quantum state fitting of measurement distributions. The contribution here is an integrated, claim-gated evidence package linking these strands with explicit boundary analysis.

4 Model and Hamiltonian decomposition

We work with staggered matter on N sites and open boundaries. The spin Hamiltonian is decomposed as

$$H(\theta) = H_{\pm} + H_{ZZ} + H_Z + H_{\text{const}}, \tag{1}$$

with

$$H_{\pm} = \frac{1}{2a} \sum_{n=0}^{N-2} \left(\sigma_n^+ \sigma_{n+1}^- + \sigma_n^- \sigma_{n+1}^+ \right) = \frac{1}{4a} \sum_{n=0}^{N-2} (X_n X_{n+1} + Y_n Y_{n+1}), \tag{2}$$

$$H_Z = \frac{m}{2} \sum_{n=0}^{N-1} (-1)^n Z_n + (\text{linear terms from electric energy}). \tag{3}$$

Using Gauss' law with background field $\alpha = \theta/(2\pi)$,

$$L_n = \alpha + \sum_{k=0}^{n} q_k, \qquad q_k = \frac{Z_k + (-1)^k}{2}, \tag{4}$$

and electric energy coefficient $\frac{g^2 a}{2}$, we obtain nonlocal ZZ couplings and additional Z shifts. The implementation computes explicit pair and linear coefficients induced by the open-boundary kernel $w_{ij} = N - 1 - \max(i, j)$.

5 Quench protocol and exact dynamics

Given (m, g, a), initial angle θ_0, and final angle θ_1, we:

1. Build $H(\theta_0)$ and compute its ground state $|\psi_0\rangle$ via exact diagonalization.
2. Evolve under $H(\theta_1)$:
$$|\psi(t)\rangle = e^{-iH(\theta_1)t} |\psi_0\rangle, \tag{5}$$
using dense eigensystem propagation for tiny Hilbert spaces and Krylov `expm_multiply` otherwise.

6 Measured observables and diagnostics

For each time point we compute:

$$\epsilon(t) = \frac{1}{N} \langle\psi(t)| H(\theta_1) |\psi(t)\rangle \,, \tag{6}$$

$$M(t) = \frac{1}{N} \sum_n (-1)^n \langle Z_n \rangle_t \,, \tag{7}$$

$$C_{ij}(t) = \langle Z_i Z_j \rangle_t - \langle Z_i \rangle_t \langle Z_j \rangle_t \,, \tag{8}$$

$$\lambda(t) = -\frac{1}{N} \log |\langle\psi_0|\psi(t)|\psi_0|\psi(t)\rangle|^2 \,, \tag{9}$$

plus bipartite von Neumann entropy $S_A(t)$ at the half-chain cut.

7 Stabilizer-Renyi magic

For each Pauli string $P \in \{I, X, Y, Z\}^{\otimes n}$ and pure state $|\psi\rangle$,

$$c_P = \langle P \rangle \,, \qquad \Xi_P = \frac{1}{d} c_P^2, \qquad d = 2^n. \tag{10}$$

We evaluate

$$M_\alpha(\psi) = \frac{1}{1-\alpha} \log \sum_P \Xi_P^\alpha - \log d, \tag{11}$$

with emphasis on $\alpha = 2$. The code streams through Pauli strings in symplectic (x, z) representation and batches parity evaluations.

8 NNQS learnability experiment

Snapshots $\{|\psi(t_k)\rangle\}$ are sampled in the computational basis from $p(z) = |\psi_z|^2$. We train three model classes to learn $p_\theta(z)$ by minimizing NLL: an autoregressive GRU, an exactly normalized RBM (small-N), and an independent Bernoulli baseline. Metrics:

- validation NLL,
- exact $\mathrm{KL}(p\|p_\theta)$ on all 2^N bitstrings,
- total variation distance,
- epochs to threshold NLL.

We regress validation NLL/KL/TV versus snapshot M_2 and apply bootstrap/permutation/meta-analysis claim gates. In the generated figure set, the primary scatter uses final validation NLL versus M_2, while exact KL and TV are retained in accompanying CSV outputs.

9 Validation checks

We enforce and report:

- Hermiticity of $H(\theta_0)$ and $H(\theta_1)$,

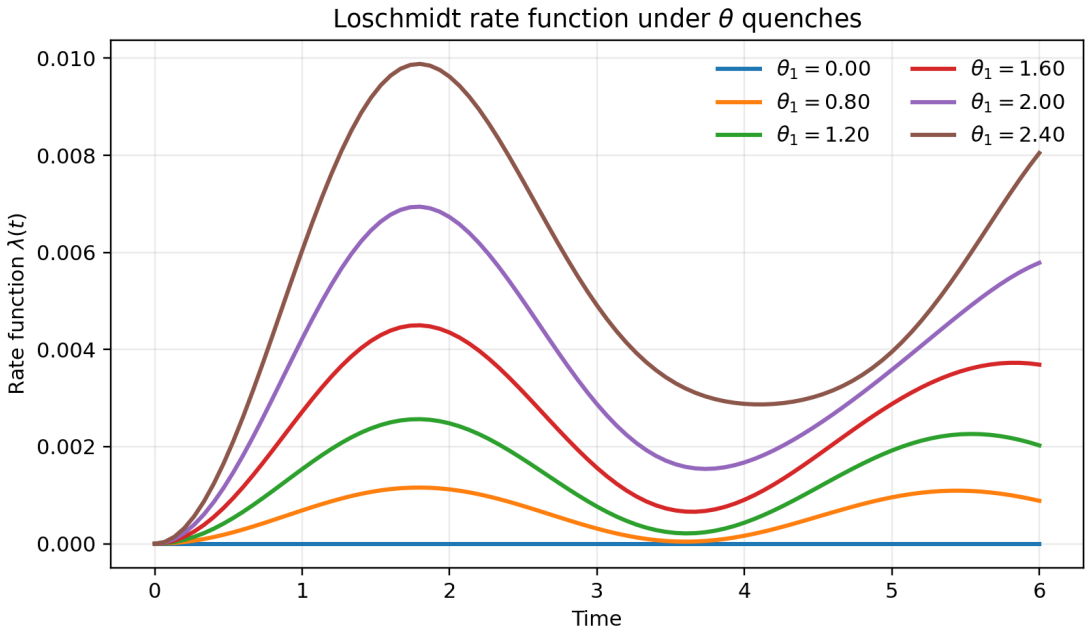

Figure 1: Loschmidt rate function $\lambda(t)$ for multiple quench endpoints θ_1. Nonanalytic-like sharp features emerge under stronger quenches.

- norm conservation drift $\max_t \left| \|\psi(t)\|^2 - 1 \right|$,
- dense-vs-Krylov trajectory mismatch for small-N,
- magic sanity states: $|0\rangle^{\otimes n}$ and GHZ (low magic), $|T\rangle^{\otimes n}$ (higher magic).

10 Figure set and interpretation

11 Evidence summary for the claim

Baseline ($N = 6, 8$) cross-architecture runs support `magic` $\rightarrow$ `validation NLL` and the entropy-controlled variant. A higher-seed full-θ rerun remains supported. However, stronger $N = 10$ exact-magic stress runs and multi-regime replications show that this relation is not universal under current settings. The defensible statement is therefore conditional: magic–learnability coupling is regime- and size-dependent in this small-N exact study.

Gate status is tracked in a standardized effect registry (`outputs/stage1_prxq/stage1/effect_registry.json`) with explicit flags for headline-region support and documented boundary failures.

11.1 Hyak checkpoint-GPU replication update

To tighten the evidence package beyond the original workstation runs, we executed additional checkpoint-GPU replications on Hyak and synced the outputs back into the repo under `outputs_hyak_runs/`. These runs do not change the qualitative claim; they make it more precise. In the main corridor, the baseline architecture story is stronger than before: GRU, MADE, and RBM all reproduce a positive magic–validation-NLL relation, and the entropy-controlled endpoint remains positive in each case. The current exact-$N = 10$ rerun remains noisy at the raw-correlation level,

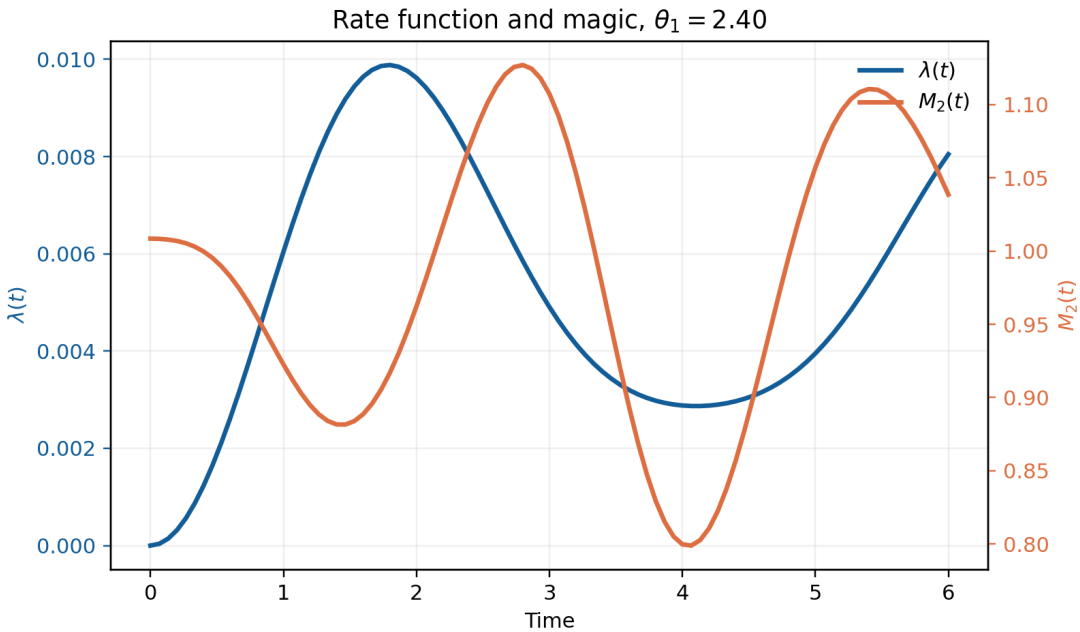

Figure 2: Overlay of $\lambda(t)$ and $M_2(t)$ for a strong quench. Peaks in transition diagnostics co-occur with rapid magic growth windows.

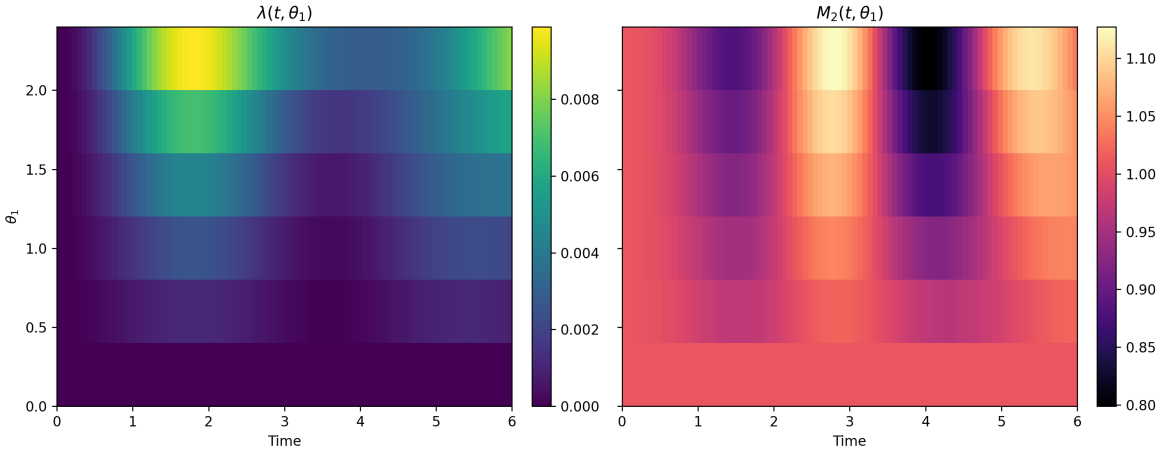

Figure 3: Heatmaps over (t, θ_1) for Loschmidt rate and magic. Quench-strength dependence appears in both landscapes.

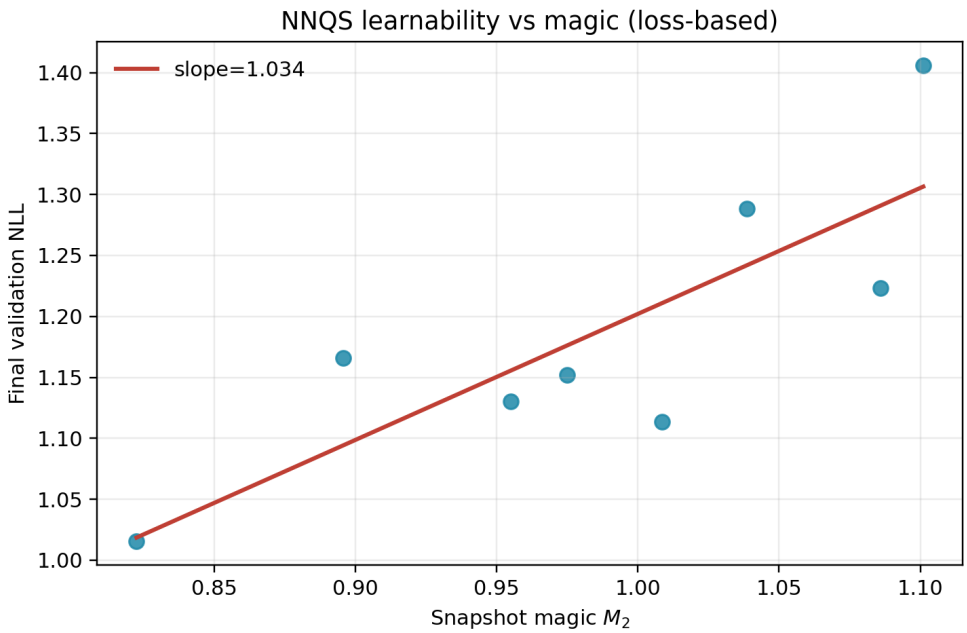

Figure 4: NNQS final validation loss versus snapshot magic M_2. The fitted slope quantifies the learnability penalty from non-stabilizerness; exact KL and TV are also reported in CSV outputs.

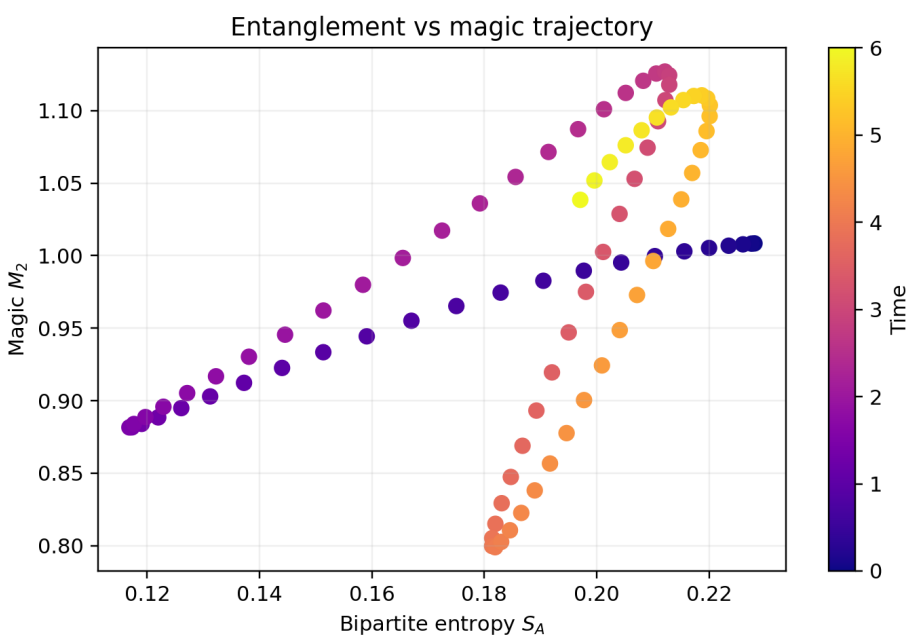

Figure 5: Optional entropy-versus-magic trajectory colored by time, showing relation and mismatch between entanglement and stabilizer complexity.

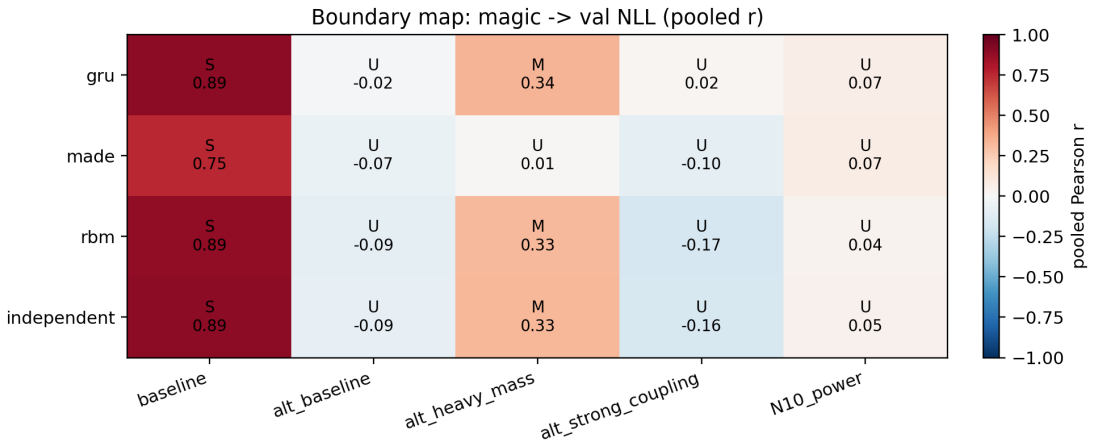

Figure 6: Regime/size boundary map for magic vs validation NLL (pooled correlation, with verdict annotations). Baseline support does not universally transfer across alternate (m, g) regimes or the current $N = 10$ exact setup.

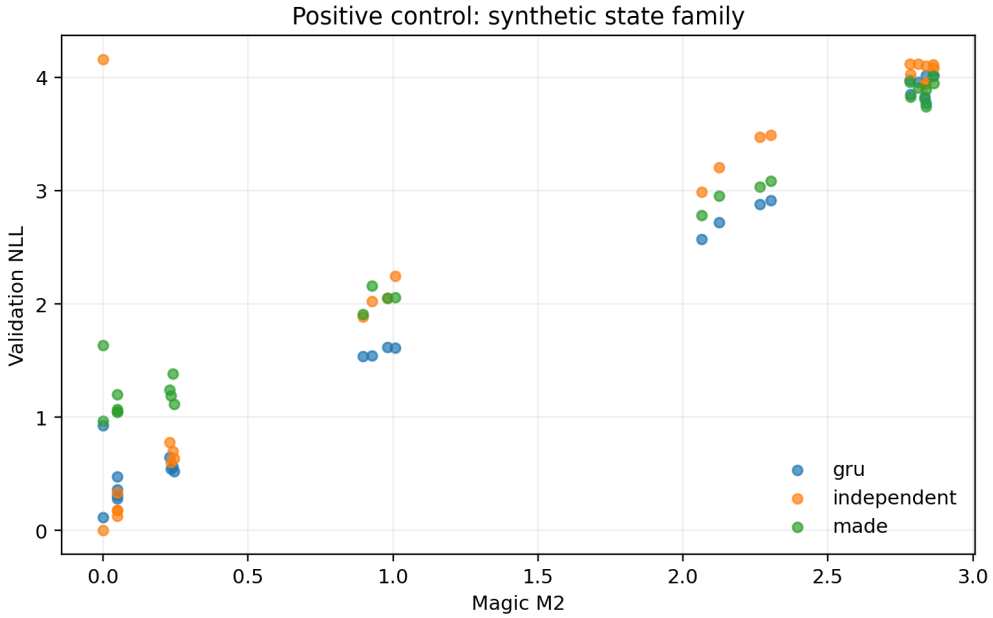

Figure 7: Positive-control synthetic family: increasing non-stabilizerness produces higher optimization difficulty (validation NLL) across NNQS baselines, supporting construct validity of the learnability metric.

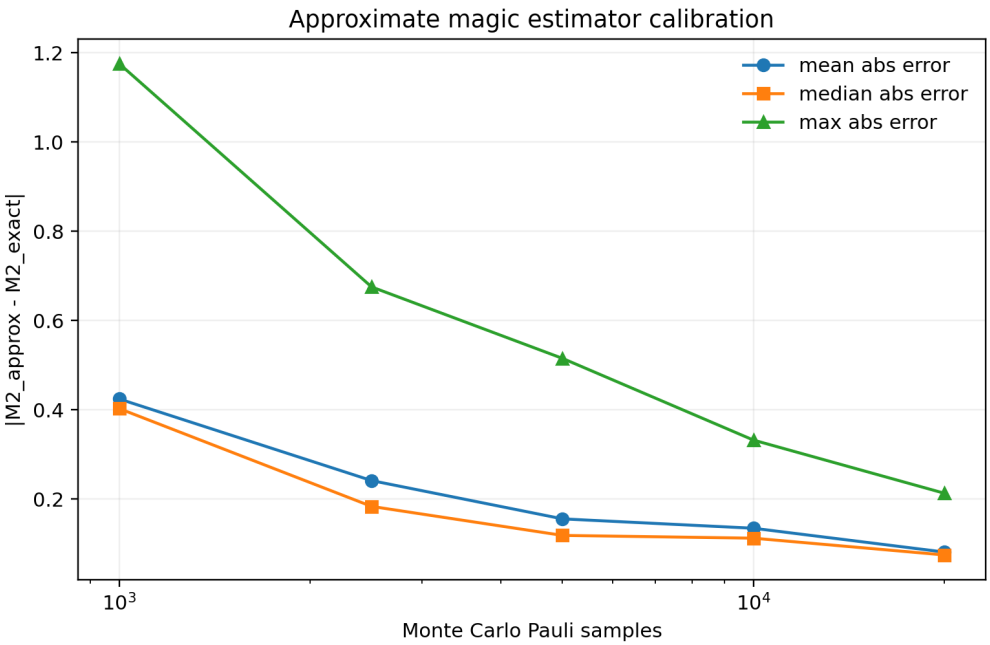

Figure 8: Approximate magic estimator calibration versus exact M_2 at tractable sizes. Error decreases with Pauli-sample budget and defines the reported approximation envelope.

but the entropy-controlled endpoint stays positive across GRU, RBM, and independent baselines. Alternate regimes sharpen the support boundary rather than extending it: heavy-mass is mixed and strong-coupling is unsupported.

12 Cross-model expansion (Schwinger + TFIM)

To stress-test generalization, we augment Schwinger outputs with a second many-body family (TFIM), and evaluate a dynamical corridor defined by large Loschmidt-rate conditions. We report:

- pooled Fisher-z meta-effects for the primary endpoint `magic` $\rightarrow$ `val NLL` | `entropy`,
- bootstrap confidence bounds for the entropy-controlled slope β_{magic} in valNLL $\sim 1 + \text{magic} + S_A$,
- finite-size trend fits versus $1/N$,
- Pinsker sanity ($\text{KL} \geq 2\,\text{TV}^2$).

The all-architecture beta-law gate passes ($\beta_{\text{magic}} > 0$ with confidence and quantile robustness), including a cluster-robust mechanism regression with positive lower bounds for each architecture. However, the all-architecture correlation-sign gate fails due heterogeneity in one weak TFIM baseline, while the expressive-only sign gate (GRU/MADE) passes across tested corridor quantiles.

13 Adversarial expansion (adding XXZ)

We then perform a stricter adversarial extension by adding XXZ quenches and testing two null-disproof gates at cell level (model family $\times$ architecture): (i) bootstrap CI lower bound on β_{magic}

Replication run	Endpoint summary	Pooled effect	Verdict
GRU baseline full-θ, $N = 6$	`magic` $\rightarrow$ `val NLL`	$r = 0.857$, $q = 1.79 \times 10^{-12}$	supported
GRU baseline full-θ, $N = 6$	`magic` $\rightarrow$ `val NLL` | `entropy`	$r = 0.913$, $q = 2.01 \times 10^{-10}$	supported
MADE, $N = 6, 8$	`magic` $\rightarrow$ `val NLL` | `entropy`	$r = 0.745$, $q = 3.10 \times 10^{-6}$	supported
RBM, $N = 6, 8$	`magic` $\rightarrow$ `val NLL` | `entropy`	$r = 0.896$, $q = 2.39 \times 10^{-18}$	supported
Exact $N = 10$ (GRU/RBM/independent)	raw `magic` $\rightarrow$ `val NLL`	opposite-sign / noisy in this rerun	unsupported
Exact $N = 10$ (GRU/RBM/independent)	`magic` $\rightarrow$ `val NLL` | `entropy`	$r \approx 0.50$–0.58	supported
Heavy-mass alternate regime	primary and entropy-controlled endpoints	positive but unstable across architectures	mixed
Strong-coupling alternate regime	primary and entropy-controlled endpoints	no stable positive effect	unsupported

Table 1: Completed Hyak checkpoint-GPU replications used to update the scoped claim. Summary files are stored under `outputs_hyak_runs/`.

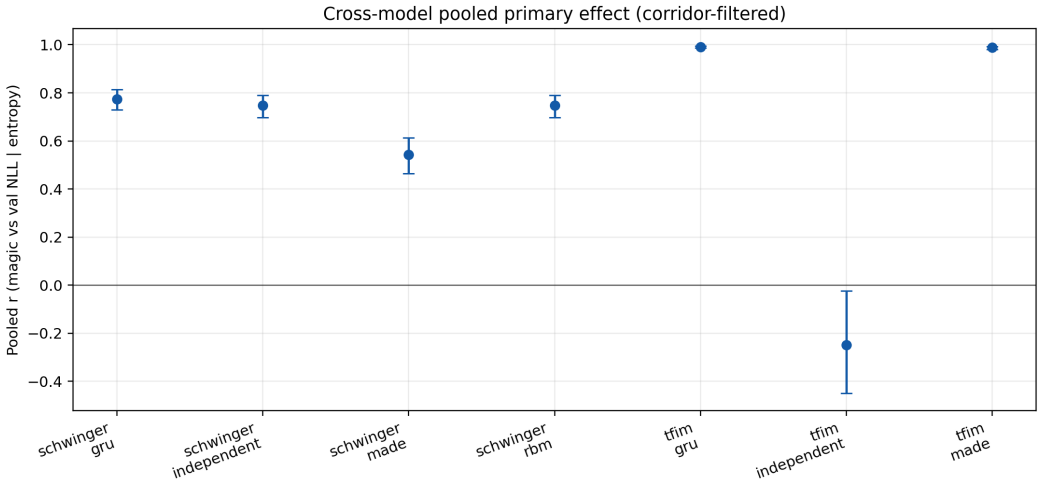

Figure 9: Stage-2 pooled primary endpoint effects (`magic` $\rightarrow$ `val NLL` | `entropy`) across Schwinger and TFIM.

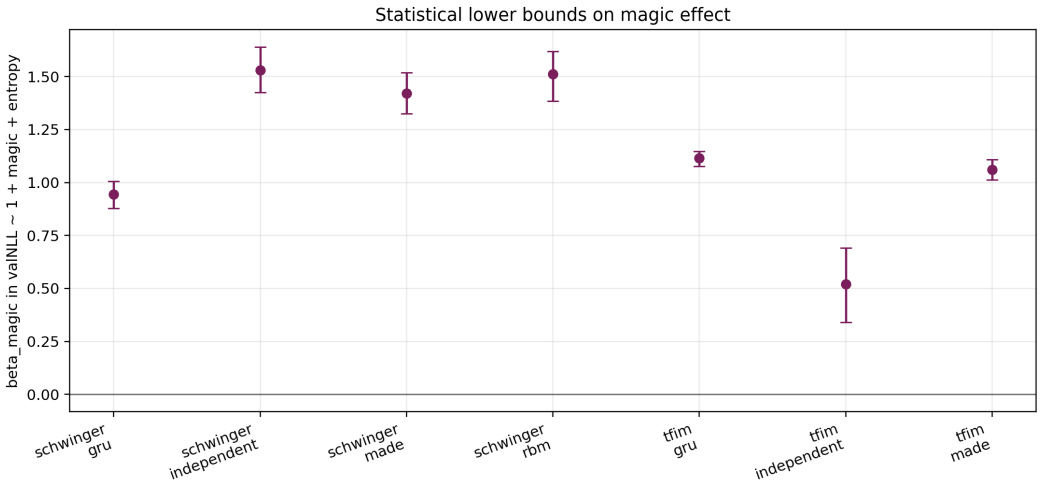

Figure 10: Bootstrap confidence intervals for entropy-controlled magic slope β_{magic}.

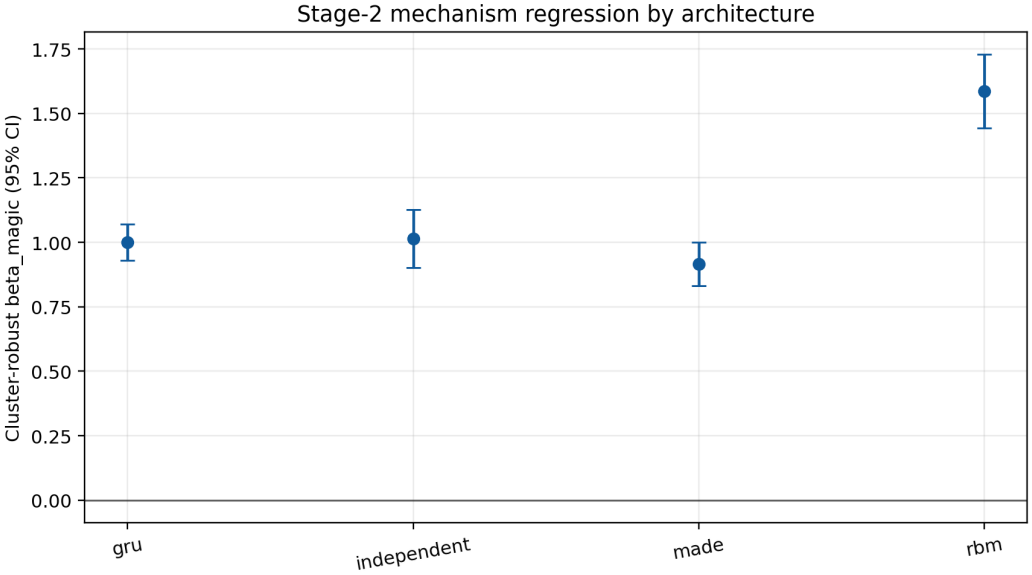

Figure 11: Cluster-robust mechanism regression: architecture-specific confidence intervals for β_{magic}.

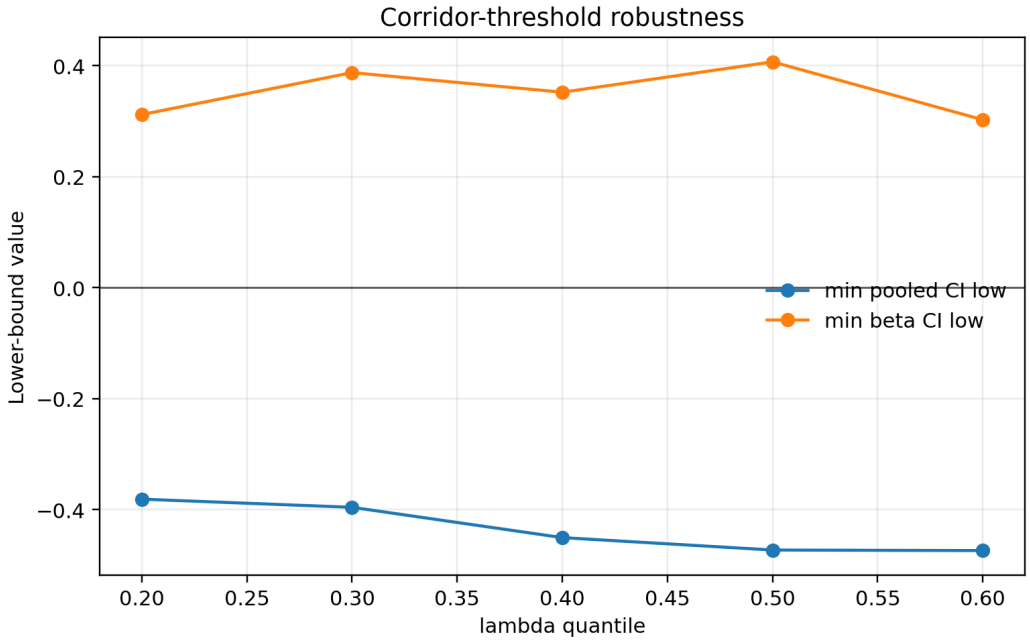

Figure 12: Corridor-threshold robustness for all architectures: beta-law remains supported across tested quantiles, while sign-law is not robust.

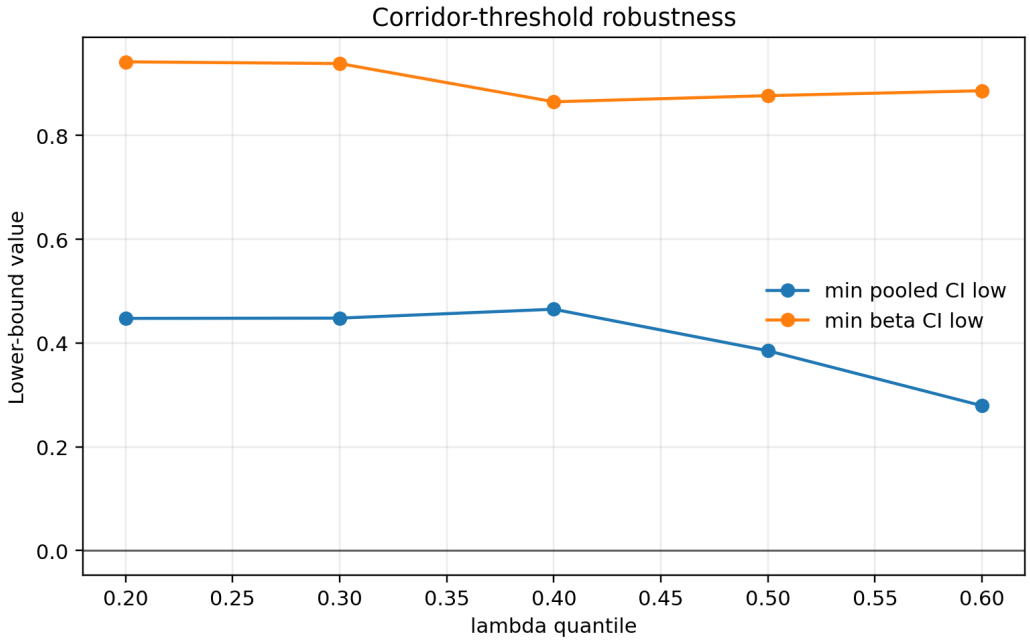

Figure 13: Corridor-threshold robustness for expressive architectures (GRU/MADE): gate passes across tested quantiles.

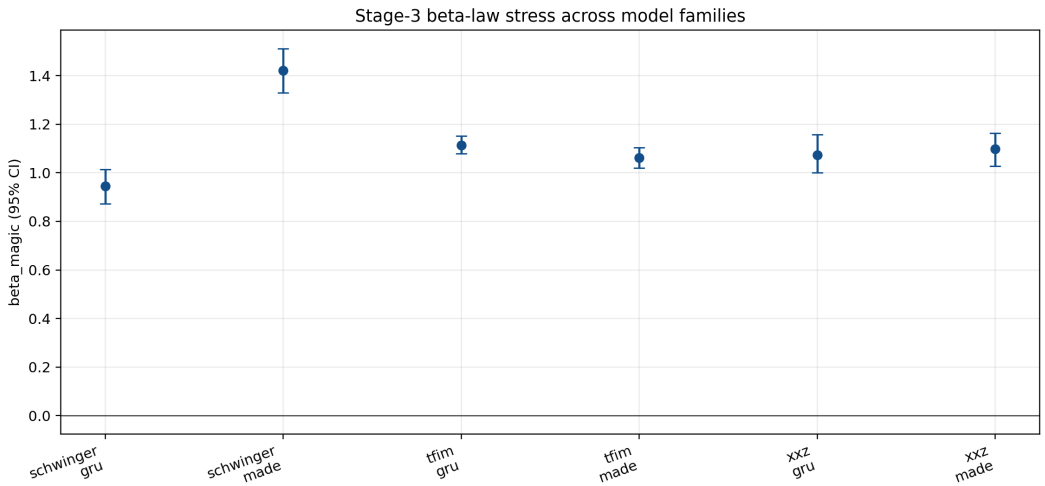

Figure 14: Stage-3 cell-level β_{magic} confidence intervals across Schwinger, TFIM, and XXZ.

and (ii) one-sided permutation null with BH-FDR correction. We include an XXZ regime tuning plus size stress through $N = 12$ using an approximate-magic calibration envelope, and evaluate the expressive architecture subset (GRU/MADE).

Result: in this tested corridor, both the extended beta-law gate and the sign-law gate pass. This strengthens the mechanistic statement for expressive models: entropy-controlled slope positivity is robust under cross-family extension, and partial-correlation direction is consistent inside the declared corridor.

14 Finite-size and lower-envelope checks

We further stress the beta-law with a finite-size extrapolation and a lower-envelope (quantile regression) check:

- finite-size cell fits of $\beta_{\text{magic}}(N)$ with weighted extrapolation to $N \to \infty$,
- quantile regression ($\tau = 0.1, 0.2, 0.3$) of validation NLL on magic and entropy controls as an empirical lower-bound proxy.

Outcome: both gates are positive in the tested corridor dataset: the all-cell thermodynamic extrapolation gate ($\beta_{\infty,\text{CI low}} > 0$) and the quantile lower-envelope gate are satisfied. This supports a stronger empirical law in the declared corridor, while remaining distinct from a theorem-level model-independent claim.

15 External-family and regime stress (adding ANNNI)

We extend the study by adding a fourth many-body family (ANNNI) with multiple additional regimes, then re-run corridor-gated beta-law and lower-envelope checks. The purpose is to test whether the mechanism survives adversarial cross-regime stress.

Outcome: family×architecture cell-level gates remain positive (bootstrap lower bounds and permutation-FDR), and quantile lower-envelope positivity remains intact. However, strict all-regime minimax positivity fails due Schwinger `alt_baseline` cells. This establishes a sharper

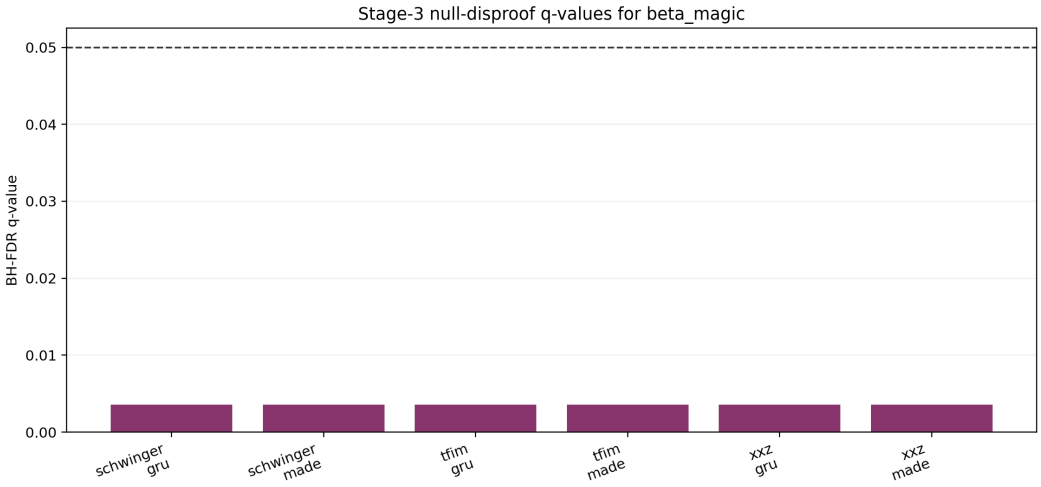

Figure 15: Stage-3 permutation null-disproof q-values (BH-FDR). In the tuned extension corridor, all beta-law cells pass $q < 0.05$.

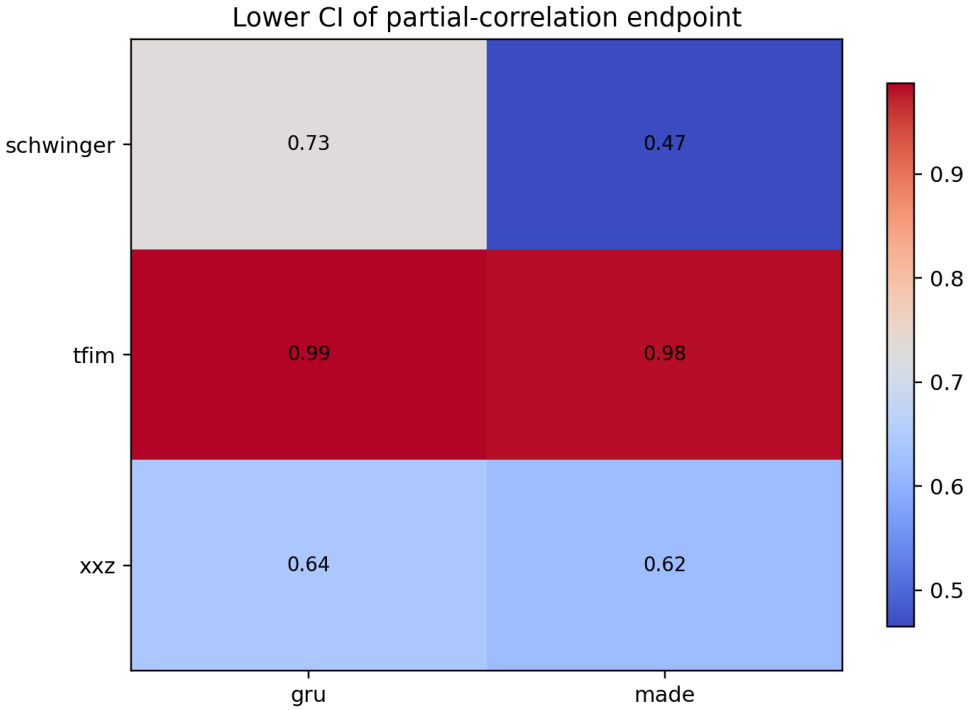

Figure 16: Stage-3 lower-CI heatmap for partial-correlation endpoint in the expressive corridor; all tested cells remain positive.

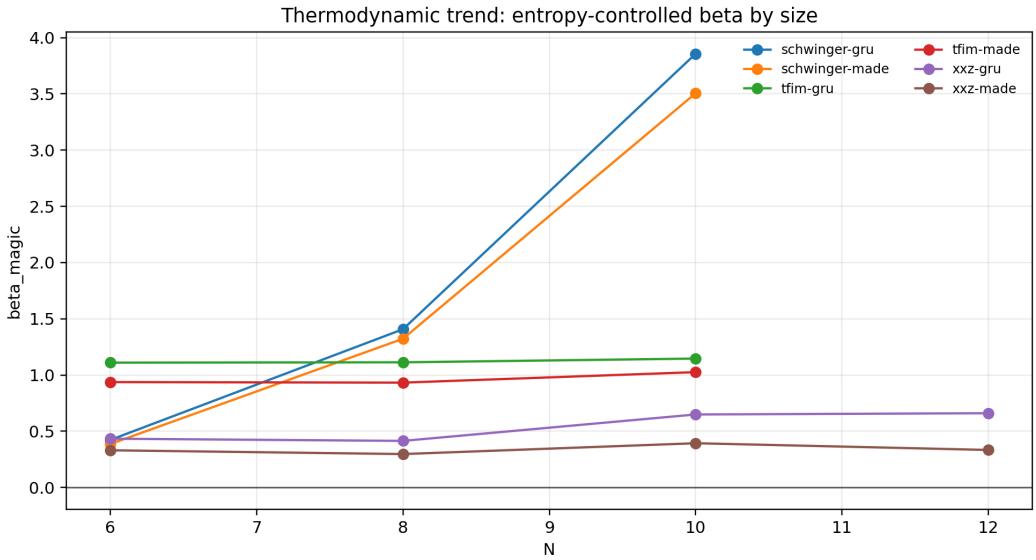

Figure 17: Stage-4 finite-size trends of β_{magic} by model family and architecture.

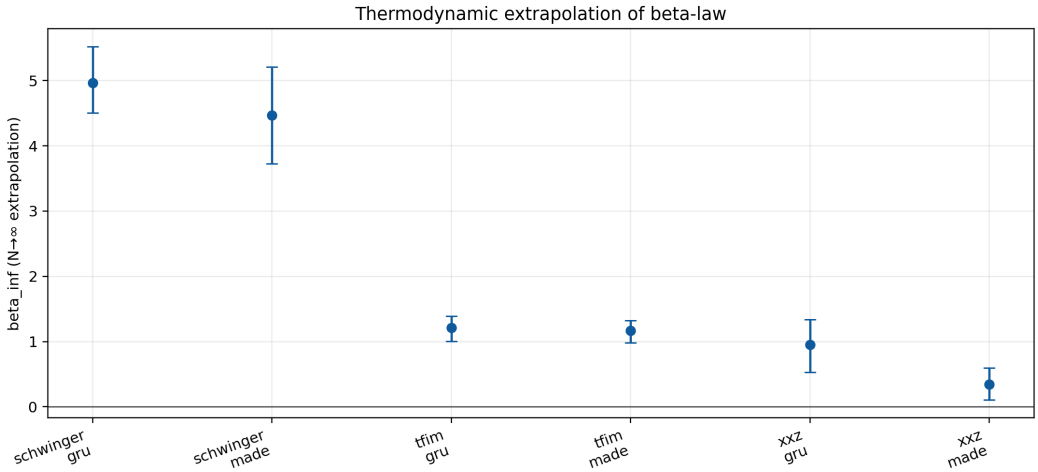

Figure 18: Stage-4 weighted extrapolation of β_{magic} to $N \to \infty$: all tested corridor cells remain positive within uncertainty.

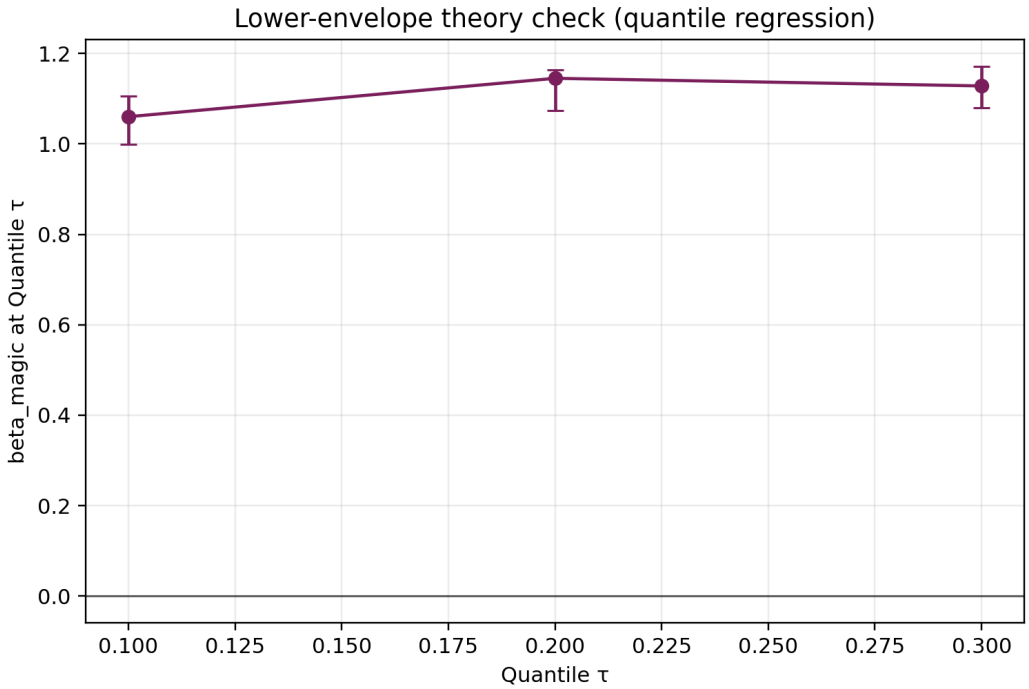

Figure 19: Stage-4 quantile-regression lower-envelope check: magic coefficient remains positive at tested quantiles.

boundary statement: the mechanism is robust across families in the declared corridor, but not globally regime-universal.

16 Ablations

We include seed, architecture, and regime ablations through the robustness suite and summarize outcomes in claim-gated reports. The primary endpoint is `magic` $\rightarrow$ `validation NLL` | `entropy`; KL/TV are treated as secondary diagnostics. Negative controls (random-label and shuffled-time baselines) are included in the condition-level statistics to detect spurious correlation structure.

We additionally include positive-control synthetic state-family tests and mixed-effects regression analyses (reported in `report/statistical_rigor.md`) to separate magic effects from entropy-only explanations.

17 Limitations and next steps

Current exact routines are restricted by Hilbert-space growth and Pauli enumeration cost. Practical next steps:

1. Use symmetry sectors and sparse block structure for larger N.
2. Replace exact magic with Monte Carlo Pauli estimators at $N \gtrsim 12$.
3. Extend NNQS architecture (Transformer/MADE) and include amplitude+phase learning from rotated measurement bases.

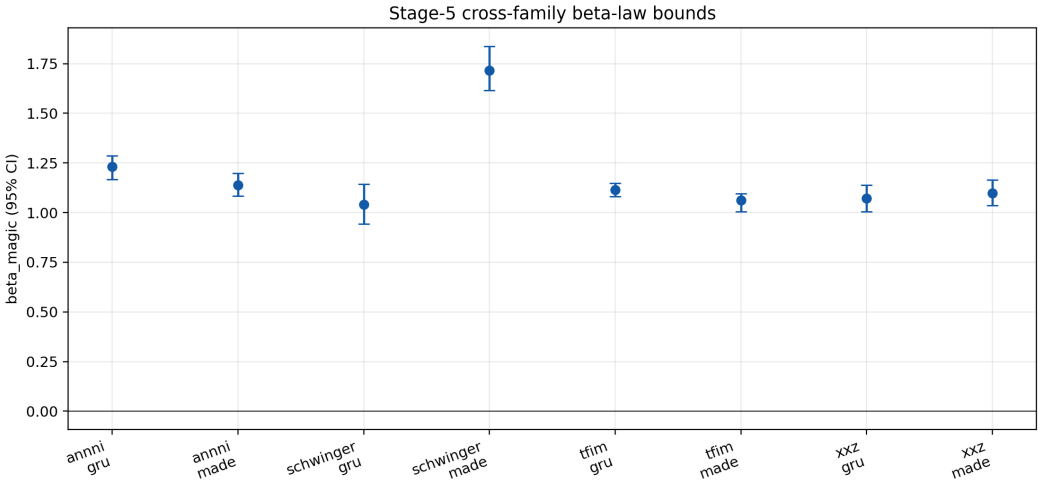

Figure 20: Stage-5 beta-law bounds after adding ANNNI. Cell-level positivity is preserved across tested families and architectures.

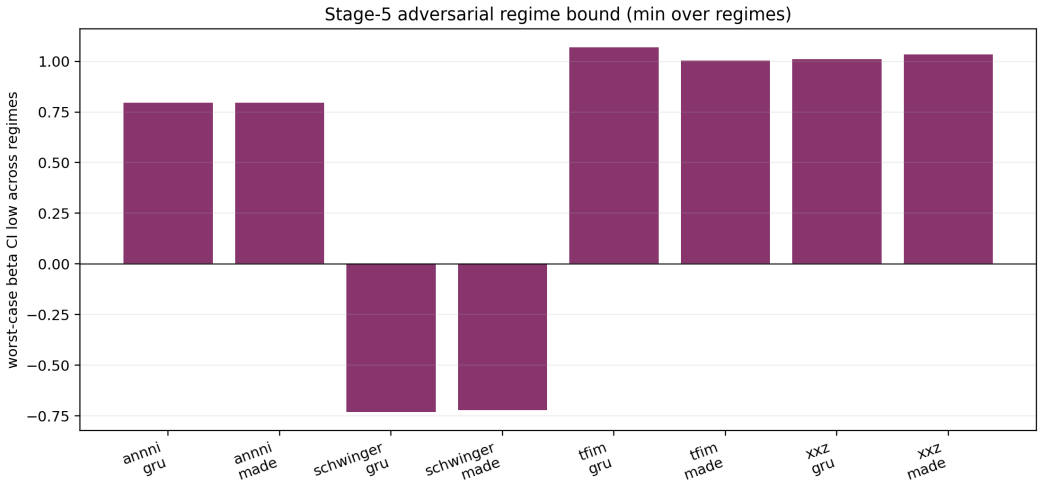

Figure 21: Stage-5 minimax regime bound: worst-case regime lower bounds identify where strict regime-universal claims fail.

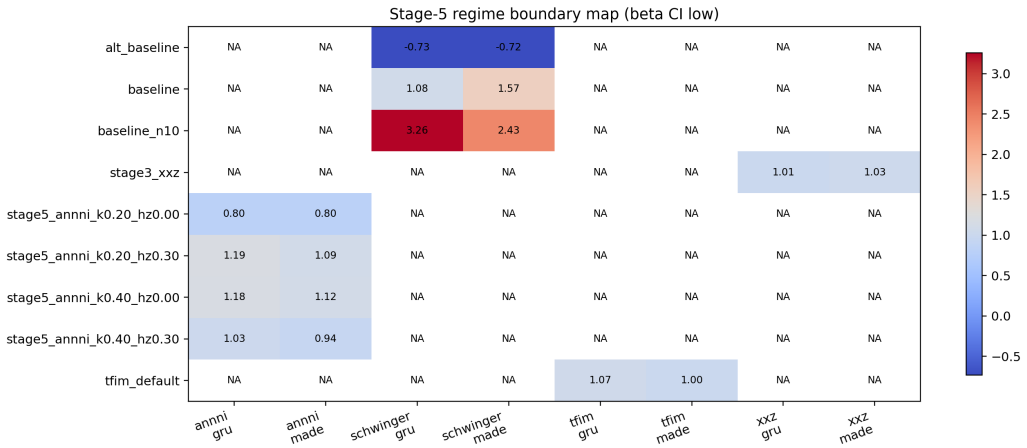

Figure 22: Stage-5 regime boundary heatmap (β_{magic} CI lower bound), explicitly marking supportive vs failing regions.

4. Add Trotterized digital-circuit compilation for direct quantum hardware execution.

18 Reproducibility

From repository root:

```
python -m tqm.run --config configs/default.yaml --all
```

This generates data in `outputs/` and figures in `outputs/figs/`. The report builder auto-syncs figure assets into `report/figs/` before compiling this manuscript. The default run includes NNQS and requires PyTorch. A torch-free fallback is available via `configs/no_nnqs.yaml`.

19 Reproducibility appendix

The repository includes:

- pre-registered analysis protocol (`report/analysis_protocol.md`),
- standardized study registry (`configs/studies/*.yaml`),
- standardized effect registry output (`outputs/<study>/<phase>/effect_registry.json`),
- Stage-2 registry and robustness packet (`outputs/stage2_prlx/stage2/effect_registry.json`, `report/stage2_completion.md`),
- camera-ready freezing command (`make camera-ready`) that archives configs, outputs, and manuscript assets.